# Automated Reference Genome Assembly in Galaxy: 152 species and counting

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Recent improvements in genome sequencing and assembly promise to generate high-quality reference genomes for many species. Yet the genome assembly process is still laborious and costly, requires substantial expertise, and is generally not scalable to the goals of multispecies scientific efforts. To democratise the training and assembly process, we implemented the latest version of the Vertebrate Genomes Project assembly pipeline in Galaxy, published earlier this year (Larivière et al. 2024).

The automated pipeline performs de novo assembly based on PacBio HiFi reads, with optional extended graph phasing using Hi-C or parental data. It supports scaffolding using Bionano optical maps data and Hi-C via modular workflows. The workflows include quality control throughout the assembly process using GenomeScope, gfastats, Merqury, BUSCO, and Pretext.

Within the Vertebrate Genome Project, these workflows have already been applied to de novo assemble genomes of over a hundred species, doubling the number of species reported in Larivière et al. 2024. We will present an overview of the new species assembled in Galaxy and what we learned from them. We will also present the updates to the published workflows and new workflows added to the VGP suite since its publication to assist with manual curation, including the integration of Treeval (Pointon, Eagles, and Sims, n.d.).

The VGP's long-term goal is to use these workflows to generate high-quality, complete reference genomes for all of the roughly 70,000 extant vertebrate species, facilitating a new era of discovery across the life sciences. We will highlight how we use new developments in Planemo (Bray et al. 2023) to accelerate the production of genome assemblies through the command line.


- Bray, Simon, John Chilton, Matthias Bernt, Nicola Soranzo, Marius van den Beek, Bérénice Batut, Helena Rasche, et al. 2023. "The Planemo Toolkit for Developing, Deploying, and Executing Scientific Data Analyses in Galaxy and beyond." Genome Research 33 (2): 261–68.
- Larivière, Delphine, Linelle Abueg, Nadolina Brajuka, Cristóbal Gallardo-Alba, Bjorn Grüning, Byung June Ko, Alex Ostrovsky, et al. 2024. "Scalable, Accessible and Reproducible Reference Genome Assembly and Evaluation in Galaxy." Nature Biotechnology, January. https://doi.org/10.1038/s41587-023-02100-3.
- Pointon, D. L., W. Eagles, and Y. Sims. n.d. "Sanger-Tol/treeval v1. 0.0–Ancient Atlantis. 2023." Publisher Full Text.

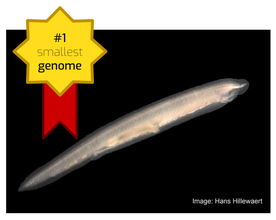

Scalable, Accessible and Reproducible Reference Genome Assembly and Evaluation in Galaxy.

## Hall of Fame

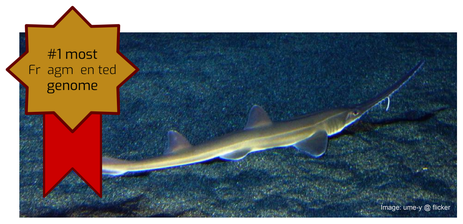
***Emys orbicularis*** **(European pond turtle)**
- Genome size: 2.32 Gbp
- 273 contigs -> 159 scaffolds
- Largest contig N50 (~85.5 Mbp, L=9)
- Scaffold N50: 137.74 Mbp (L=5)

***Gavialis gangeticus*** **(gharial)**
- Genome size: 2.33 Gbp
- 278 contigs -> 220 scaffolds
- Second largest contig N50 (~84.2 Mbp, L=12)
- Scaffold N50: 255.85 Mbp (L=5)

***Pseudophryne corroboree*** **(Corroboree frog)**
- Genome size: 8.88 Gbp
  - That's almost 3x human!
- 5,841 contigs -> 3,315 scaffolds
- Contig N50: 6.82 Mbp, L=353
- Scaffold N50: 792.63 Mbp, L=5
- Up to 3cm in body length (FrogID)




***Branchiostoma lanceolatum*** **(lancelet)**
- Invertebrate chordate
- Genome size: 473.22 Mbp
- 5,841 contigs ->132 scaffolds
- Contig N50: 16.38 Mbp, L=13
- Scaffold N50: 23.14 Mbp, L=7
- Pictured example is ~22mm




***Pristiophorus japonicus*** **(Japanese sawshark)**
Genome size: ~5.5 Gbp
- Hap1: 11,071 contigs → 4,429 scaffolds
  - N50: 1.86 Mbp (L=867) → 206.1 Mbp (L=10)
- Hap2: 12,149 contigs → 5,698 scaffolds
  - N50: 1.85 Mbp (L=921) → 192.4 Mbp (L=12)
- Troublesome sample for the wet lab: 7 SMRT cells → ~50x covg
- At least half the HiFi data was considered short (peaked just under 8 Kbp, cf. goal of ~15 Kbp)

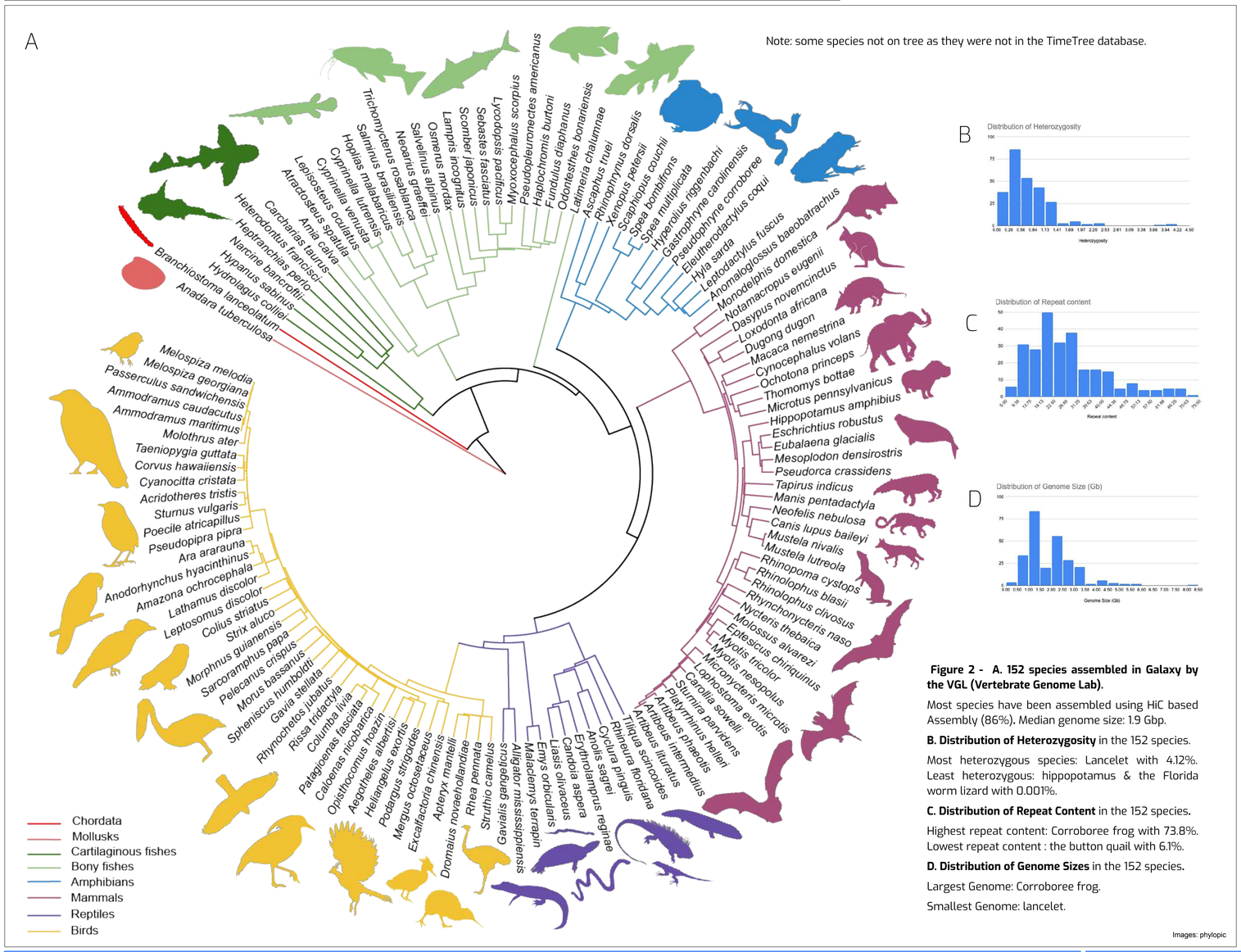



**Figure 2 - A. 152 species assembled in Galaxy by the VGL (Vertebrate Genome Lab).**

Most species have been assembled using HiC based Assembly (86%). Median genome size: 1.9 Gbp.

**B. Distribution of Heterozygosity** in the 152 species.

Most heterozygous species: Lancelet with 4.12%. Least heterozygous: hippopotamus & the Florida worm lizard with 0.001%.

**C. Distribution of Repeat Content** in the 152 species.

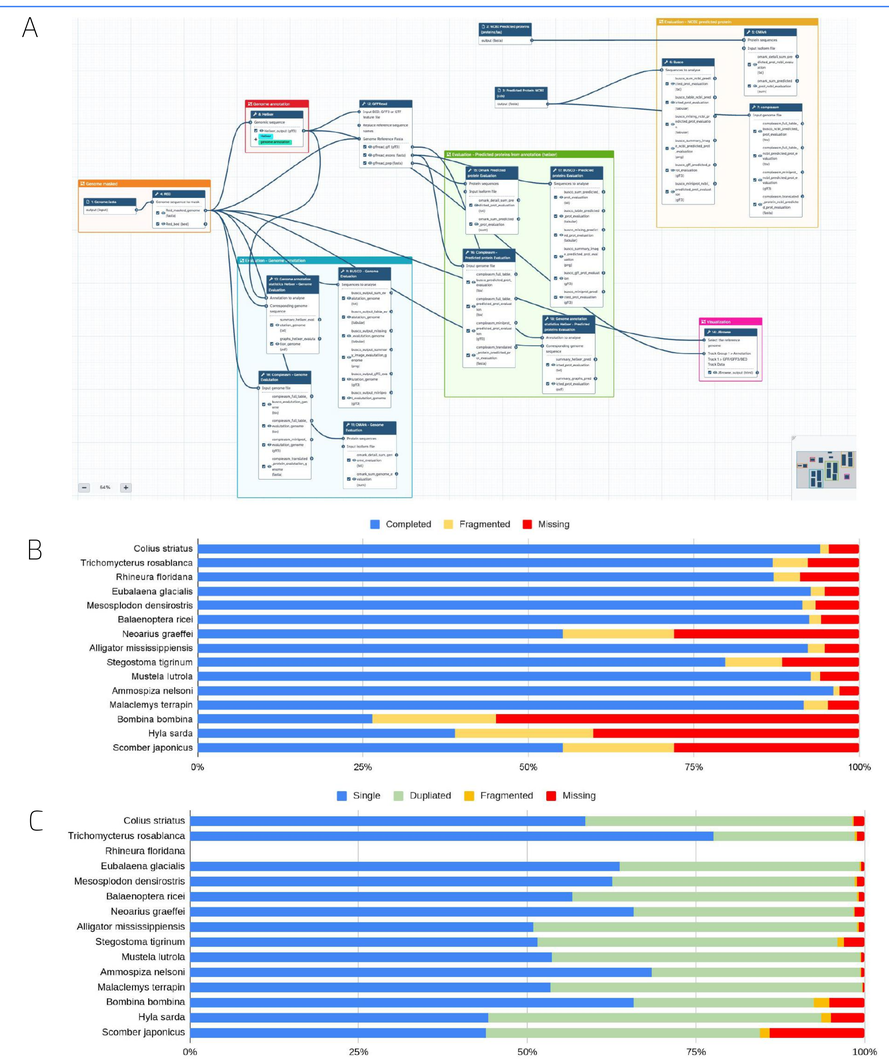
Highest repeat content: Corroboree frog with 73.8%. Lowest repeat content : the button quail with 6.1%.

**D. Distribution of Genome Sizes** in the 152 species.

Largest Genome: Corroboree frog.

Smallest Genome: lancelet.

Images: phylopic




Romane Libouban

Anthony Bretaudeau

SEE POSTER "Community-driven standards development for reference genome generation" *Tom Brown et al.*

**Figure 3 - Annotation in Galaxy with Helixer. A. Annotation Workflow in Galaxy.** Orange: Genome masking using Red or RepeatMasker. Red: Genome Annotation with Helixer. Blue: Evaluation of Genome Annotation. Green: Evaluation of protein prediction. QC: BUSCO, Compleasm, OMArk and Genome annotation statistics. Pink: Visualization with JBrowse. Yellow: Comparison with NCBI annotation. **B,C. Annotation Benchmarking of Helixer (B) and NCBI (C) Annotations with Compleasm on 15 VGP assembled species**.

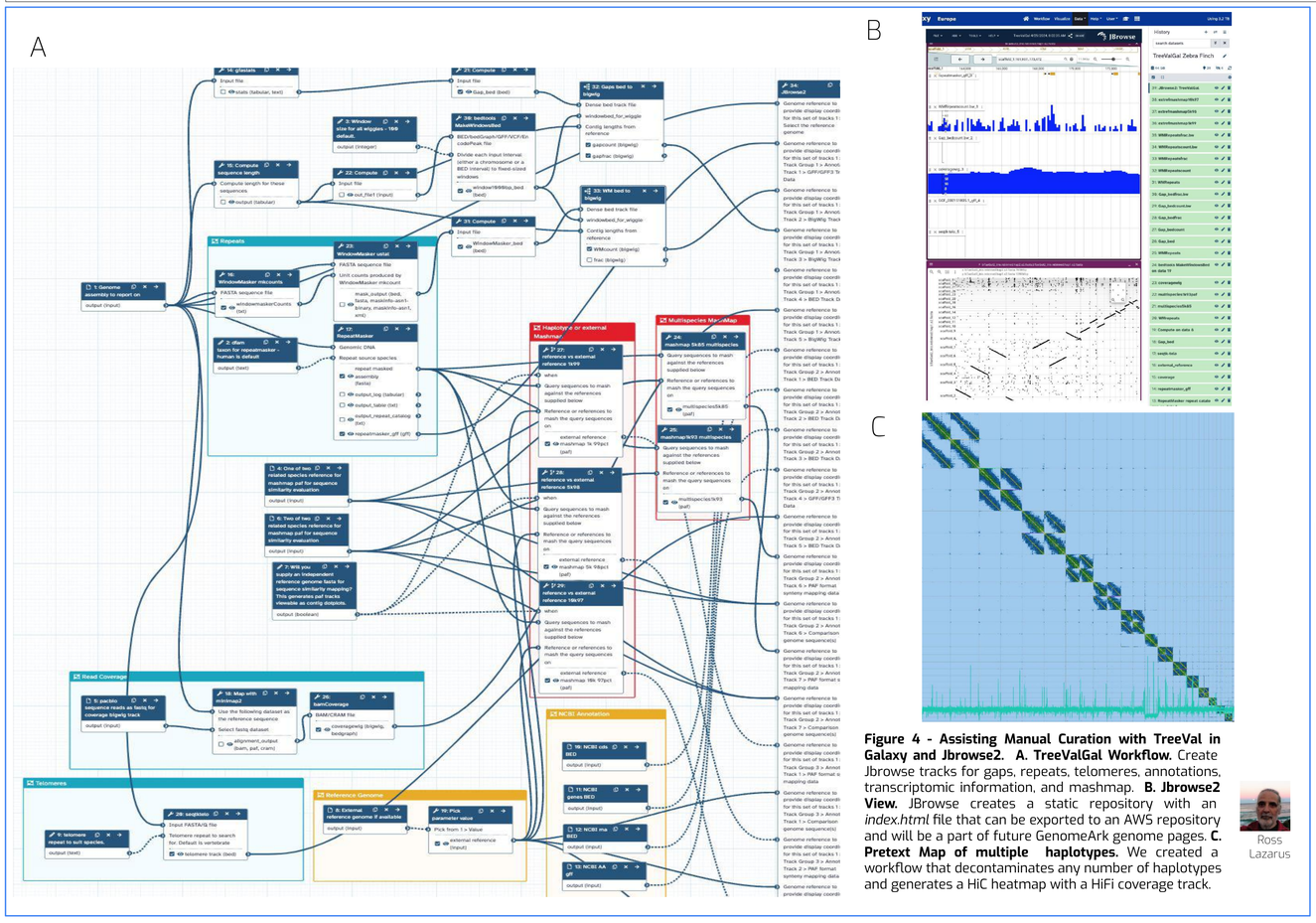



**Figure 4 - Assisting Manual Curation with TreeVal in Galaxy and Jbrowse2. A. TreeValGal Workflow.** Create Jbrowse tracks for gaps, repeats, telomeres, annotations, transcriptomic information, and mashmap. **B. Jbrowse2 View.** JBrowse creates a static repository with an *index.html* file that can be exported to an AWS repository and will be a part of future GenomeArk genome pages. **C. Pretext Map of multiple haplotypes.** We created a workflow that decontaminates any number of haplotypes and generates a HiC heatmap with a HiFi coverage track.

Ross Lazarus

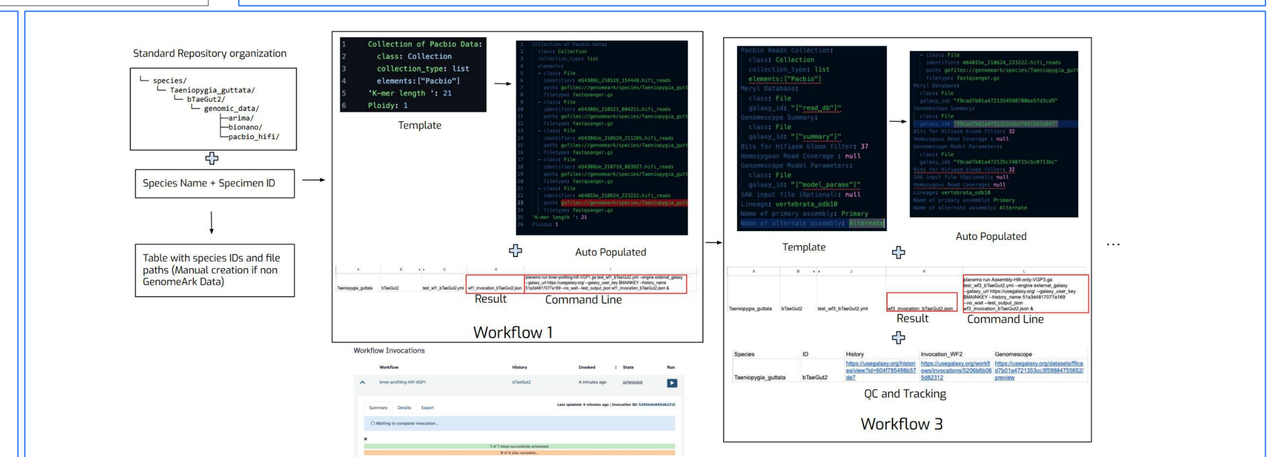



**Figure 5 - Accelerating the submission to free public resources by command line with Planemo.** We created scripts to auto-populate job files with GenomeArk paths and outputs from previous workflows. We keep track of metadata and QC in tables updated at each step.

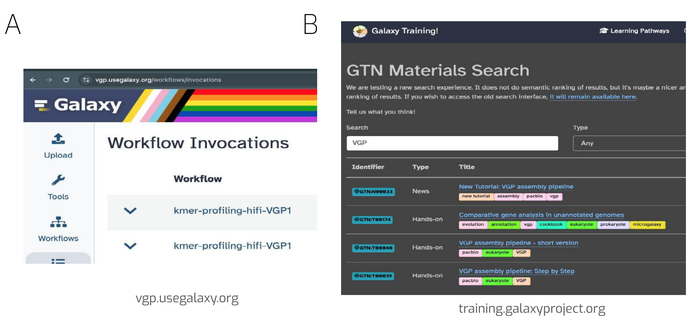



**Figure 6 - A community dedicated to accessible Genome Assembly. A. VGP Instance with dedicated resources. B.- Training Material on Genome assembly, simple and advanced.**